

UMM ARE THERE ANY LIMITATION?

They made a big “food book” using meals from many countries. A huge chunk of that food book comes from the U.S. so the paper very good on american food right? So dont it confused about other foods ?

The paper kinda admits this in a smart way . They don't say “US dominance” directly like a headline, but they show it by saying: Data comes from multiple countries BUT heavily includes U.S. dataset .Models perform better on some countries, worse on others . They say we need more diverse cultural data .

ENHANCING HALLUCINATION DETECTION THROUGH NOISE INJECTION

What I get from abstract..

the robot sounds very confident but gives a wrong answer. That's called a hallucination. So the problem is We want to catch when the robot is wrong.

So people try this: Ask the robot the same question many times ,If it gives different answers each time, it's probably confused → maybe wrong . here's the issue

Normally, the robot just: icks slightly different words each time

But that doesn't fully show how unsure it really is.

The new idea (this paper)

Instead of just asking again...

They shake the robot's brain a little while it's thinking

Not breaking it — just adding a tiny bit of randomness (called noise)

Why does this help?

- If the robot really knows the answer:
 - Even if you shake it → it still gives similar answers
- If the robot is guessing / hallucinating:
 - After shaking → answers become very different

So the trick is:

1. Ask the question many times
 2. Slightly “shake” the model each time
 3. Check:
 - Are answers similar → probably correct
 - Are answers all over the place → probably wrong
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So the Big takeaway is

Shaking the model helps reveal when it's unsure

And that makes it easier to detect fake or wrong answers

Core idea: hallucination = LLM says something confidently wrong. Detection = measure answer variation.

Step 1: What is a hallucination?

Step 2: What is “detection”? We want to catch when the AI is wrong.

Step 3: The trick — ask multiple times

Step 4: Measure “variation” (how much answers change)

Low variation (good sign)

High variation (bad sign)

It's not just:

“wrong answer = hallucination”

It's:

“unstable answers = likely hallucination”

Even if:

- Sometimes it accidentally says the correct answer
- But keeps changing → still suspicious

